



Riemannian and Complex Geometry

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— Exercises —

7.1 Homework for mini course by Song Sun

Exercise 7.1. Let V be a real vector space of dimension 4 and let μ be a nonzero element in $\Lambda^4 V$. Define a symmetric bilinear form Q on $\Lambda^2 V$ by

$$Q(\alpha, \beta) := \frac{\alpha \wedge \beta}{\mu}.$$

Prove that Q has signature $(3, 3)$.

Exercise 7.2. Let V be a real vector space of dimension 4 and let μ be a nonzero element in $\Lambda^4 V$. Suppose we are given 3 elements $\omega_1, \omega_2, \omega_3$ in $\Lambda^2 V$ such that

$$\frac{1}{2} \omega_i \wedge \omega_j = \delta_{ij} \mu, \quad 1 \leq i, j \leq 3.$$

Prove that there is a basis $\{e_1, e_2, e_3, e_4\}$ of V such that

$$\begin{cases} \omega_1 = e_1 \wedge e_2 + e_3 \wedge e_4 \\ \omega_2 = e_1 \wedge e_3 + e_4 \wedge e_2 \\ \omega_3 = e_1 \wedge e_4 + e_2 \wedge e_3. \end{cases}$$

Then ω_i spans the self-dual subspace Λ^+ with respect to the inner product g such that $\{e_i\}$ is an oriented orthonormal basis.

Exercise 7.3. Let (M, g) be a 4 dimensional oriented Riemannian manifold. Show that the Weyl curvature W , as a self-adjoint endomorphism of $\Lambda^2 = \Lambda^+ \oplus \Lambda^-$, decomposes as

$$W^\pm : \Lambda^\pm \rightarrow \Lambda^\pm,$$

and both W^+ and W^- are trace free.

Exercise 7.4. Prove that a Kähler metric ω has harmonic Ricci form if and only if it has constant scalar curvature.

Exercise 7.5. Suppose X is compact Kähler manifold and ω and ω' are two Kähler metrics with $[\omega] = [\omega'] \in H^2(X; \mathbb{R})$ and $\omega^n = (\omega')^n$. Prove that $\omega' = \omega$.

7.2 Kähler geometry

Exercise 7.6. Let (X, h) be a Kähler manifold of dimension n . Then around each point there exists a holomorphic coordinate $(z^1, \dots, z^n)$ such that

$$h_{i\bar{j}}(z) = \delta_{i\bar{j}} - \Theta_{i\bar{j}k\bar{\ell}}(p) z^k \bar{z}^\ell + O(|z|^2).$$

Exercise 7.7. Let L be a complex line bundle on a complex manifold. Prove that for any $m \in \mathbb{Z}_{>0}$ and a Hermitian metric h_m on $L^{\otimes m}$, it defines a Hermitian metric on L .

Exercise 7.8. Show that for compact Kähler manifold (X, ω) of dimension n , we have

$$\begin{aligned} [\Delta_d, L] &= 0 \\ [L, \Lambda] &= (k - n) \text{id} \quad \text{on } C^\infty(X, \Omega_{X, \mathbb{C}}^k). \end{aligned}$$